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MicroRNA Expression Abnormalities in Limited Cutaneous Scleroderma and Diffuse Cutaneous Scleroderma

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Abstract Scleroderma (systemic sclerosis, SSc) is a complex autoimmune disease caused by progressive fibrotic replacement of normal tissue architecture, a progressive and ultimately fatal process that currently has no cure. Although dysregulation of microRNAs (miRNAs) is known to be involved in a variety of pathophysiologic processes, the role of miRNAs in SSc is unclear. In comparison with the normal skin tissues, miRNAs were aberrantly expressed in limited cutaneous scleroderma and diffuse cutaneous scleroderma skin tissues. We also identified miRNAs whose expressions were correlated with SSc fibrosis: miR-21, miR-31, miR-146, miR-503, miR-145, and miR-29b were predicted to be involved. This study further confirmed that miR-21 was increased whereas miR-145 and miR-29b were decreased both in the skin tissues and fibroblasts. As predicted target genes, SMAD7, SAMD3, and COL1A1 were regulated by these miRNAs. After stimulation with transforming growth factor β, the expression of miR-21 was increased and that of SMAD7 mRNA was decreased. MiR-145 was upregulated whereas the mRNA level of SMAD3 was downregulated. The downregulation of miR-29b was correlated with the upregulation of COL1A1 mRNA. MiRNAs might play an important role in the pathogenesis of SSc and suggest a potential therapy.

Keywords Scleroderma · microarray · microRNA · target gene

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Introduction

Scleroderma (systemic sclerosis, SSc) is a complex, heterogeneous, and autoimmune disease of unknown etiopathogenesis, characterized by vascular dysfunction and excessive accumulation of extracellular matrix, resulting in progressive fibrotic replacement of normal tissue architecture [1]. There are two major subgroups in the commonly accepted classification of SSc: limited cutaneous scleroderma (lSSc) and diffuse cutaneous scleroderma (dSSc) [2].

Many important cellular and molecular mechanisms and mediators of pathological fibrogenesis have been elucidated, and transforming growth factor β (TGF-β) is identified as playing a fundamental role in the process [3, 4]. Perturbed TGF-β expression and function are implicated in the pathogenesis of fibrosis in SSc [5]. Elevated levels of TGF-β have been observed in SSc skin biopsies [6, 7], and increased expression of TGF-β-regulated genes has been identified in SSc fibroblasts [8]. The canonical SMAD pathway is uniquely associated with TGF-β signaling and is deregulated in SSc [9], so TGF-β is the molecular target for antifibrotic therapy. Multiple platforms for blocking TGF-β exist in the pipeline and have a major impact on the treatment of SSc.

MicroRNAs (miRNAs) comprise a large family of ~21-nucleotide-long RNAs, which bind to the 3′untranslated regions of target genes and thereby repress the translation of target genes and/or induce the degradation of target gene mRNA [10]. MiRNAs have been shown to play fundamental roles in diverse biological and pathological processes, including intracellular signaling pathways and organ morphogenesis [11]. Aberrant expression of miRNAs is closely associated with the initiation and progression of pathophysiologic processes, and some miRNAs participate in the fibrotic process in a number of organs, including the heart, kidney, liver, and lung [12, 13]. However, the role of miRNAs in SSc is just beginning to unravel [14, 15]. Therefore determining the roles of specific miRNAs involved in the pathogenesis of fibrosis (especially in the TGF-β signaling pathway) helps to understand the pathogenesis of SSc more clearly and to identify novel methods for the diagnosis, treatment, and prevention of SSc.

Materials and Methods

Patients, Biopsy Specimens, and Cell Culture

All 14 subjects signed the informed consent approved by the local ethics committee. Patients met the classification criteria for SSc of American College of Rheumatology [16] and were further characterized as the diffuse or limited subset [2]. Seven patients underwent one set of biopsy (Table I). One 1-cm punch biopsy from the lateral forearm, 8 cm proximal to the ulnar malleolus, was taken for clinically involved skin. Seven normal controls underwent the biopsy in the identical locations (forearms). Then the biopsy was quartered, two pieces were immediately frozen, the third piece went into 10% formalin for routine histology processing, and the fourth for fibroblast cell culture.

Skin fibroblasts were expanded by outgrowth culture at 37°C in a humidified atmosphere of 95% air–5% CO_2 in Dulbecco's modified Eagle's medium (DMEM; Gibco Invitrogen) supplemented with 10% heat-inactivated fetal bovine serum (Hangzhou Sijiqing Biological Technology, China). Cells from passages 3–8 were used for the experiments. After culturing under an 80–90% confluent condition, the cells were ready for the experiments. After 24 h in the serum-reduced DMEM (0.5% fetal calf serum), healthy dermal fibroblasts were stimulated for 48 h with TGF-β (10 ng/ml) (R&D Systems).

MiRNA Microarray Analysis

Samples were sent to Beijing CapitalBio Corporation (China) for processing. MiRNAs were enriched from total RNA with mirVana® miRNA Isolation Kit (Ambion, Foster City, CA, USA) and labeled with FlashTag™ Biotin RNA Labeling Kit. Labeled miRNAs were used for hybridization on each miRNA microarray (Affymetrix). Then, images of the miRNA microarrays were acquired by Affymetrix ® GeneChip® Scanner 3000. The figure signals were transformed to digital signals using image analysis software (LuxScan3.0; CapitalBio), with the free miRNA QC Tool software for data summarization, normalization, and quality control.

Computational Prediction of miRNA Target Genes

MiRNA target genes in SSc were screened by merging the results of several computational prediction algorithms provided at the miRanda (http://microrna.sanger.ac.uk), TargetScan (http://www.targetscan.org), and PicTar (http://pictar.bio.nyu.edu).

RNA Isolation and Quantitative Real-Time PCR Analysis of miRNA

Total RNA from the skin biopsy specimens and fibroblasts was isolated with the miRNeasy mini kit (Qiagen).The miScript reverse transcription kit (Qiagen) and miScript SYBR Green PCR kit (Qiagen) were used to measure the expression levels of selected miRNAs in a model 7500 real-time PCR system analyzer (Applied Biosystems), with a miRNA-specific primer and the miScript Universal Primer (Qiagen). Expression of the U6B small nuclear RNA (RNU6B) was used as the endogenous control to normalize the sample data. To determine the quantity of mRNA, cDNA was prepared by Reverse Transcription System (Promega, USA) and amplified by real-time PCR with SYBR Green (SYBR Premix Ex Taq RT-PCR kit; Takara, Shiga, Japan) with the primers. Expression of the GAPDH was used as the endogenous control to normalize the sample data. Relative expression levels were calculated with the $2^{-\Delta\Delta Ct}$ method. The following primers were used: GAPDH forward: GCACCGTCAAGGCTGAGAAC, GAPDH reverse: TGGTGAAGACGCCAGTGG A; SMAD7 forward: TGCTGTGCAAAGTGTTCAGGTG, SMAD7 reverse: CCATC GGGTATCTGGAGTAAGGA; SMAD3 forward: AGGCGTGCGGCTCTACTACAT C, SMAD3 reverse: CAGCGAACTCCTGGTTGTTGAA; and COL1A1 forward: TCTAG ACATGTTCAGCTTTGTGGAC, COL1A1 reverse: TCTGTACGCAGGTGATTGG TG.

Immunohistochemistry

The skin biopsy specimens were inflated with 10% formalin solution and embedded in paraffin. Some skin sections, 5 μm thick, were prepared and stained with hematoxylin and eosin for histological examination; some others were deparaffinized with xylene and rehydrated in water through graded ethanol. Antigen retrieval was performed in a pressure cooker in Tris–EDTA solution, pH 9.0, for 5 min. After incubation in TBST buffer for 10 min and 3% H_2O_2 for 10 min, the sections were

Table I Subjects clinical characteristics

Subject	Age/ sex	Duration, mos	Skin score	ANA/Scl-70	Organs	Medications
dSSc 1	30/F	12	24/51	+(1:160 speckled)/+	RP	CTX, penicillamine
dSSc 2	64/F	2	9/51	+(1:160 speckled)/+	RP	prednisone, penicillamine
dSSc 3	62/M	2	31/51	+(1:160 nucleolar)/+	RP, lung	CTX, prednisone
dSSc 4	60/M	6	47/51	+(1:320 speckled)/+	Lung	CTX, penicillamine
dSSc 5	45/F	6	33/51	+(1:180 speckled)/+	RP, lung	MTX, colchicine
lSSc 1	38/F	36	11/51	+(1:320 speckled)/+	RP	MTX, penicillamine
lSSc 2	44/F	48	13/51	+(1:180 speckled)/+	RP, lung	MTX, penicillamine
Normal 1	39/F	–	–	–	–	–
Normal 2	49/M	–	–	–	–	–
Normal 3	21/M	–	–	–	–	–
Normal 4	58/M	–	–	–	–	–
Normal 5	51/F	–	–	–	–	–
Normal 6	36/F	–	–	–	–	–
Normal 7	58/M	–	–	–	–	–

In each case, the age, sex, and duration of disease as defined by the onset of first non-Raynaud's phenomenon symptom, modified Rodman skin scores on a 51-point scale, ANA or anti-Scl-70 titers, internal organ involvement, and medication taken by each patient are indicated.

M male, *F* female, *RP* Raynaud's phenomenon, *MTX* methotrexate, *CTX* cyclophosphamide, *ANA* anti-nuclear antibody

washed with TBST, blocked with goat serum, and then incubated with rabbit anti-human SMAD3 and SMAD7 antibody (Abcam) overnight at 4°C and then secondary antibody for 45 min, avidin–horseradish peroxidase for 45 min, and DAB chromagen sequentially. Finally, the sections were counterstained with hematoxylin.

Statistical Analysis

GraphPad Prism software was used for statistical analysis. Numerical variables with normal distribution were compared with unpaired *t* test or paired *t* test. Non-normal distribution data were compared with Wilcoxon rank sum test. Data were expressed as mean ± SEM. $P<0.05$ was considered statistically significant.

Results

A miRNA Expression Signature Discriminated SSc Tissues from the Normal Skin

In SSc patients, an abnormal accumulation of extracellular matrix constituents, most notably collagen types I and III, was the most prominent pathological manifestation of the disease in the skin. Skin lesions were characterized by the presence of thickened collagen bundles within the reticular dermis that ran parallel to the skin surface and inflammatory cells infiltrated between the collagen bundles and around blood vessels (Fig. 1a). We used a custom microarray platform validated by numerous studies to evaluate the miRNA expression profiles on a heterogeneous set of skin tissues from different patients. This set included three normal skin samples, four dSSc skin samples, and two lSSc skin samples, a total of nine biologically independent samples. Each skin sample was derived from a single specimen. To identify the differentiation of normal versus SSc tissues by miRNAs, we used SAM tool, and the SAM comparison between the normal and the SSc tissues identified different miRNA signatures. Among the 847 individual miRNAs represented on the microarrays, 42 miRNAs were significantly regulated in the dSSc skin biopsies (25 miRNAs showed an expression increase ≥1.5-fold and 17 miRNAs showed a ≥1.5-fold expression decrease). In addition, 60 miRNAs changed in the lSSc skin biopsies (42 miRNAs showed an expression increase ≥1.5-fold and 18 miRNAs showed a ≥1.5-fold expression decrease). A total of 21 miRNAs altered similarly both in dSSc and lSSc (13 miRNAs were upregulated and eight miRNA were downregulated) (Fig. 1b).

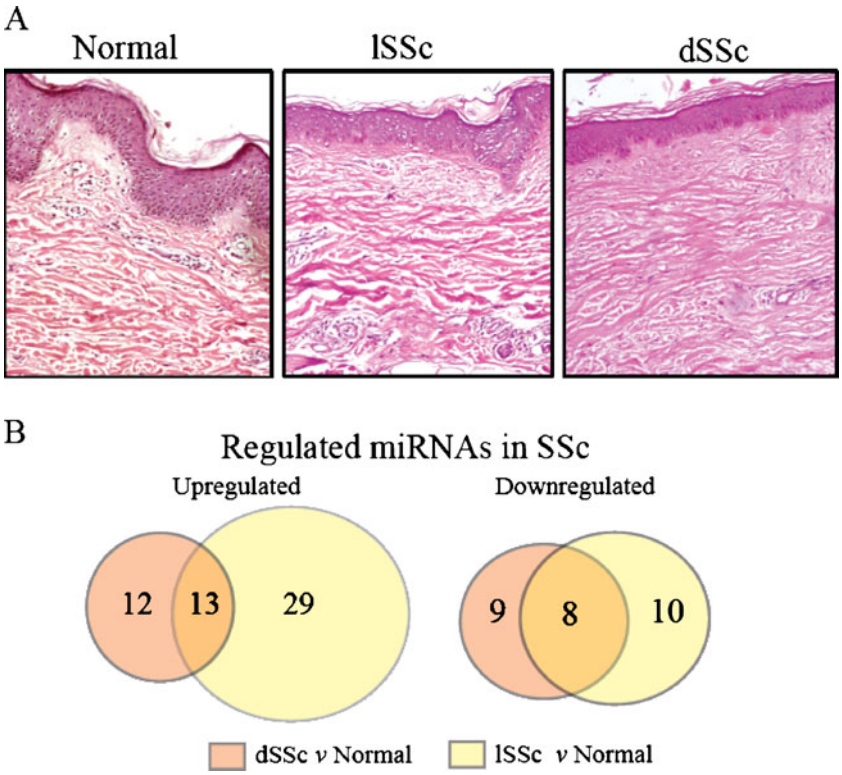

Fig. 1 MiRNA expression in SSc skin biopsies. **a** Histopathology of skin lesions in SSc characterized by an excessive deposition of collagens, deep fibrosis, and perivascular lymphohistiocytic infiltrates. **b** Microarray analysis revealed that miRNAs were dynamically regulated in dSSc and lSSc, and the number of miRNAs regulated ≥1.5-fold in each category was shown

Fibrotic Pathways and miRNA Expression

To define the possible function of altered miRNAs in SSc, we used computational predictions to identify possible targets. Correlations between each miRNA and the expression of its predicted targets, including the number of binding sites in the respective mRNAs and the variation of the correlation coefficients, were statistically analyzed. The Targetscan prediction program indicated an unexpectedly high number of fibrosis-related mRNAs encoding collagens, metallopeptidases, and integrins as possible targets for six miRNAs, namely miR-21, miR-31, miR-503, miR-146, miR-145, and miR-29b (www.targetscan.org). To determine whether the six miRNAs might regulate fibrosis, we focused on predicted targets implicated in the TGF-β signaling pathway. Collagen type

Table II Computational prediction analysis of miRNA targeting genes

ProbeSetID	Target protein	Predicted consequential pairing of target region (top) and miRNA (bottom)	Seed match
hsa-miR-21	SMAD7	5' ...UGUUUAGACUUUAACAUAAGCUA... 3' AGUUGUAGUCAGACUAUUCGAU	8mer
hsa-miR-503	SMAD7	5' ...ACUUCAAACUACUUU-GCUGCUAA... 3' GACGUCUUGACAAGGGCGACGAU	7mer-1A
hsa-miR-145	SMAD3	5' ...AGUCUAAAUUAUUUCAACUGGAA... 3' UCCCUAAGGACCCUUUUGACCUG	8mer
hsa-miR-146	SMAD4	5' ...UUUUAAAGGCAGAGAAGUUCUCA... 3' UUGGGUACCUUAAGUCAAGAGU	8mer
hsa-miR-29b	COL1A1	5' ...CCAUUUUAUACCAAAGGUGCUAC... 3' UUGUGACUAAAGUUUA----CCACGAU	7mer-1A
hsa-miR-31	Proliferation		

Sequence alignments for regulated miRNAs and their putative human mRNA targets

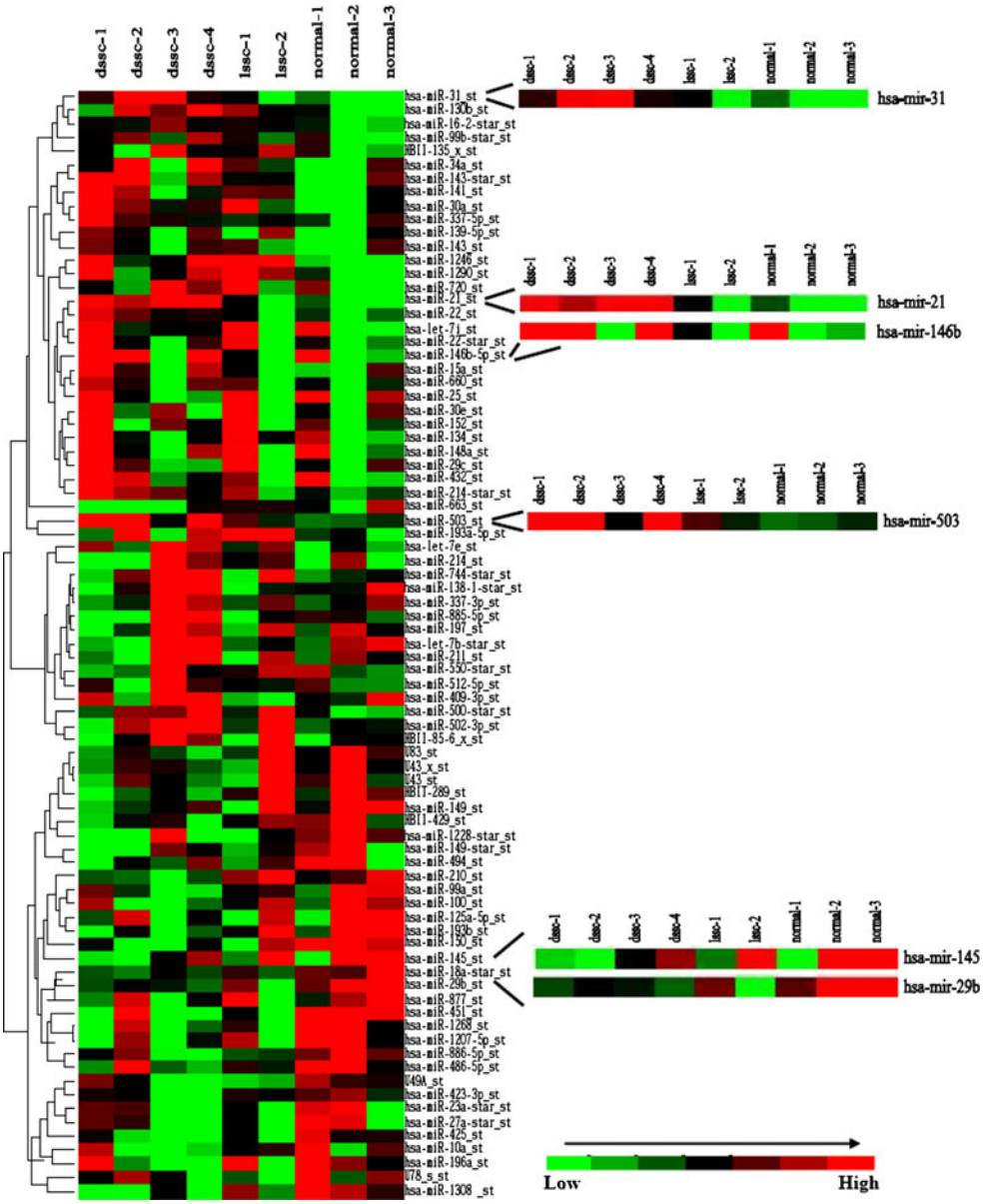

Fig. 2 Unsupervised hierarchical clustering analysis of three normal skin tissues and six SSc skin tissues (four dSSc and two lSSc). MiR-21, miR-31, miR-146, and miR-503 upregulated in SSc versus the normal tissues (red); MiR-145 and miR-29b downregulated in SSc versus the normal tissues (*green*)

I, alpha 1 and 2 (COL1A1, COL1A2) and collagen type III, alpha 1 (COL3A1), SMAD3, SMAD4, and SMAD7 all contained one or more conserved potential seed sequences for these miRNAs (Table II; Fig. 2).

Identification of SSc miRNAs by Real-Time PCR

To confirm the results from microarray analysis, we used real-time PCR for the six miRNAs involved in the fibrotic pathways and three miRNAs were identified. Expression of miR-21 significantly increased in SSc skin biopsies (Fig. 3a), whereas that of miR-145 and miR-29b was significantly downregulated (Fig. 3c, e). By isolating the skin fibroblasts, miR-21, miR-145, and miR-29b were regulated similarly in the fibroblasts (Fig. 3b, d, f).

Confirmed Potential Targets for miRNA Identification

Expression of SMAD7 in SSc Patients

As miRNAs downregulated the steady-state levels and the translation of their target mRNAs, we analyzed the expression of SMAD7 (predicted miR-21 mRNA target) in the tissue extracts isolated from the skin sections and fibroblasts. As shown in Fig. 4a, decreased levels of SMAD7 mRNA were observed in SSc patients compared with the normal controls. In the fibroblasts from the SSc patient, this was paralleled by decreased levels of SMAD7 mRNA compared with healthy control fibroblasts (Fig. 4b). Next, we investigated the expression of the SMAD7 protein in the dermis of the SSc skin biopsy samples. SMAD7 protein was downregulated in accordance with mRNA expression (Fig. 4c).

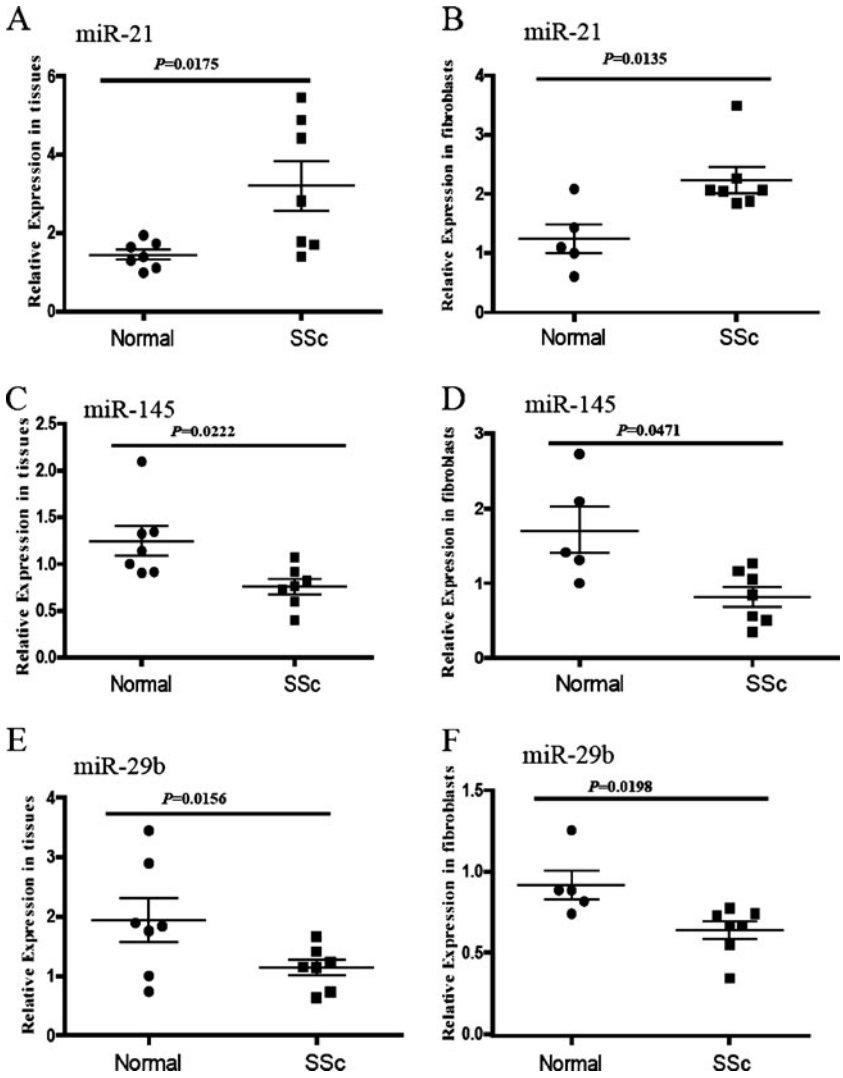

Fig. 3 Real-time PCR confirmed the regulation of specific miRNAs in the skin tissues and fibroblasts from SSc patients. MiR-21 was upregulated (**a**, **b**) while miR-145 (**c**, **d**) and miR-29b (**e**, **f**) were downregulated. Experiments were repeated at least three times

Expression of SMAD3 in SSc Patients

As SMAD3 mRNA was the predicted target of miR-145, we investigated the expression of SMAD3 mRNA in the skin samples from SSc patients and normal controls. The expression of SMAD3 mRNA in the tissue extracts isolated from the skin sections was analyzed, and the levels of the SMAD3 mRNA were increased in SSc patients (Fig. 4d). In the fibroblasts from SSc patients, the levels of SMAD3 mRNA were increased similarly compared with those of healthy control fibroblasts (Fig. 4e). The expression of SMAD3 protein in the dermis of the skin biopsy was also upregulated (Fig. 4f).

Expression of COL1A1 in SSc Patients

As a predicted target of miR-29b, the levels of COL1A1 mRNA were also detected by real-time PCR. The expression of COL1A1 mRNA in the skin tissues from SSc patients was significantly increased (Fig. 4g). Similarly, the levels of the COL1A1 mRNA were also increased in fibroblasts from SSc patients compared with those from healthy controls (Fig. 4h).

Altered Expression of miRNAs and mRNAs After TGF-β Stimulation

TGF-β has been shown to play a dominant role in the production and deposition of collagens. Real-time PCR on fibroblasts exposed to TGF-β revealed an increase in miR-21 expression and a decrease in SMAD7 mRNA (Fig. 5a). MiR-145 was upregulated significantly upon TGF-β treatment, whereas the mRNA level of SMAD3 was downregulated (Fig. 5b). The downregulation of miR-29b correlated with the upregulation of COL1A1 (Fig. 5c).

Discussion

This study revealed a collection of miRNAs that were dysregulated in the skin tissues from SSc patients. There were

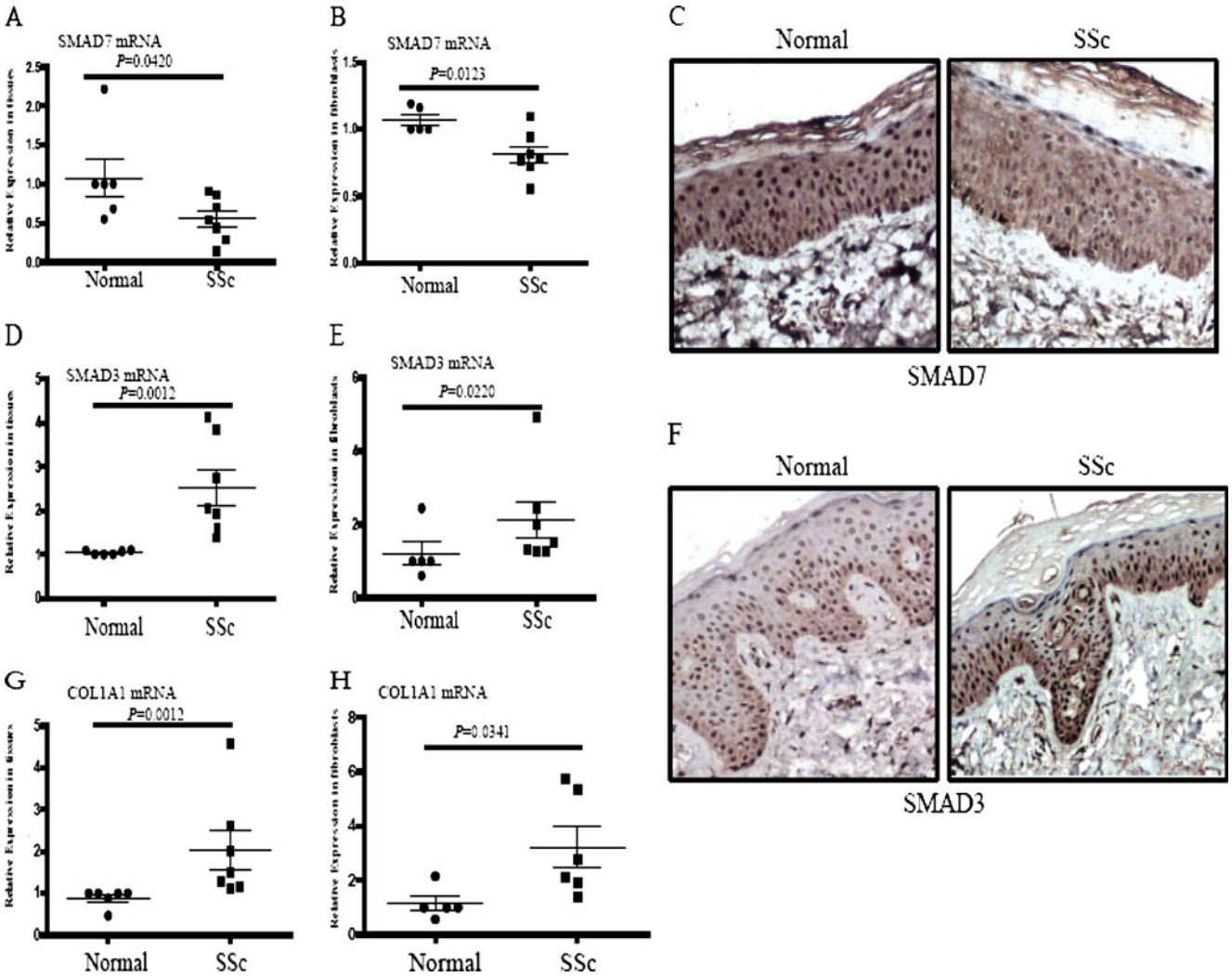

Fig. 4 MiRNA regulated target protein mRNAs: (1) Real-time PCR for the expression of predicted target genes both in the skin tissues and fibroblasts from SSc patients showed an expression decrease in SMAD7 (**a**, **b**) and an expression increase in SMAD3 (**d**, **e**) and collagens (COL1A1) (**g**, **h**) and (2) immunohistochemistry for the skin biopsy showed downregulation of SMAD7 (**c**) and upregulation of SMAD3 (**f**). Experiments were repeated at least three times

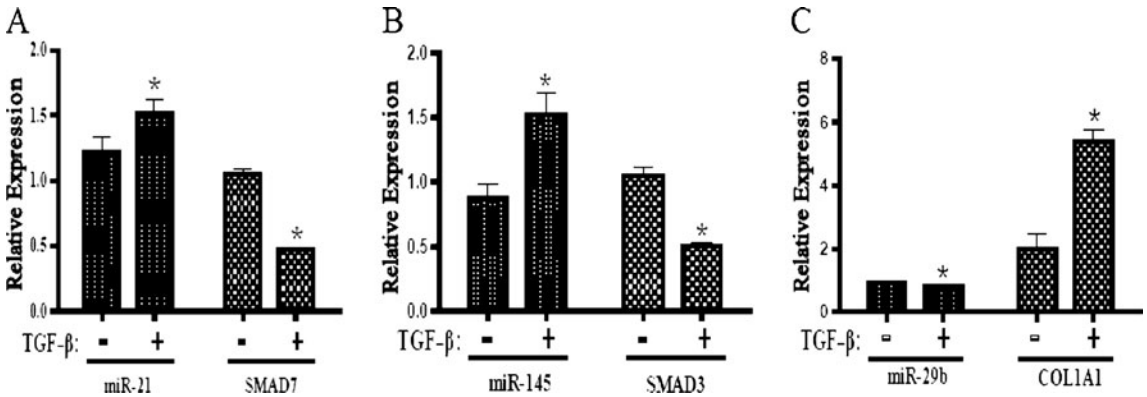

Fig. 5 Altered expression of miRNAs and mRNAs after TGF-β stimulation. Real-time PCR for fibroblasts exposed to TGF-β (10 ng/ml, 48 h) revealed an increase in miR-21 expression and a decrease in SMAD7 mRNA (**a**), an increase in miR-145 and a decrease in SMAD3 (**b**), and a decrease in miR-29b and an increase in COL1A1 (**c**). *$P<0.05$ compared with the untreated fibroblasts. Experiments were repeated at least three times

different expressions of miRNAs in dSSc and lSSc. We identified 42 miRNAs that were differentially expressed in dSSc and 60 miRNAs with altered expression in lSSc. A total of 21 miRNAs altered similarly both in dSSc and lSSc. Since dSSc and lSSc have different pathogenesis, genome-wide gene expression profiling of skin biopsies demonstrated that different gene expression patterns in lSSc and dSSc. dSSc displayed a "diffuse proliferation" gene expression profile and a "TGF-β-responsive gene signature," and lSSc displayed an "inflammatory and limited gene signature" [17]. Different regulations of miRNAs tend to contribute to many aspects of the disease response.

Among these changed miRNAs, many were predicted to function as inhibitors of numerous mRNAs involved in autoimmune, vascular, and fibrotic process, several of which were shown to be involved in the fibrotic pathway [13, 15]. Our study showed that many fibrosis-related mRNAs encoding collagens, metallopeptidases, and integrins were possible targets for the 6 miRNAs: miR-21, miR-31, miR-503, and miR-146 were upregulated; miR-29b and miR-145 were downregulated. The predicated mRNA targets (SMAD7, SMAD3, and COL1A1) were also dysregulated both in the skin tissues and fibrosis. We further stimulated fibroblasts with recombinant TGF-β which led to a significant increase in miR-21 expression and a decrease in SMAD7 expression. MiR-145 was upregulated whereas the mRNA levels of SMAD3 were downregulated, and the downregulation of miR-29b was correlated with the upregulation of COL1A1.

Although our results suggested important roles of miR-21, miR-145, and miR-29b in the controlling SSc fibrosis, there was no direct one-to-one relationship between the miRNA level and target mRNA expression. To confirm the function of altered miRNAs and determine whether these miRNAs directly regulated the target mRNA, gain-and-loss of function experiments and luciferase reporter assay to analyze their cellular origin and function in SSc are needed. Excessive TGF-β activity is a common feature for a number of fibrotic conditions of diverse etiologies [18]. The development of selective therapies blocking the fibrotic pathways is currently considered to be one of the most promising novel therapeutic approaches in SSc. Growing evidence suggests a few treatment options for fibrosis. Modulation of miRNAs using either antisense strands that block their activity or overexpression from viral or plasmid vectors has been a potentially therapy [19, 20]. MiRNA-based therapeutic approaches in animal studies of other diseases have been encouraging, since toxic effects were not observed [21–23]. Previous studies demonstrate that miR-29 family is a key regulator for collagen expression in systemic sclerosis and suggest miR-29a as a potential therapeutic target [15]. Serum levels of miR-29a levels may be useful for the differentiation of SSc from scleroderma spectrum disorder [24]. In this study, we identified three miRNAs involved in the fibrotic pathways. Further investigating the function and mechanisms of action of these specific miRNAs is likely to suggest important new directions for the treatment of SSc, and identification specific miRNAs as biomarkers of disease severity will pave the way for the development of disease-specific and stage-specific targeted therapies.

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